

PORT ISABEL CAMERON COUNTY AP, TX, USA

WMO: 722508

Lat: 26.166N

Lon: 97.346W

Elev: 19

StdP: 14.69

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 12957

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	38.9	42.9	25.8	19.8	49.3	30.9	25.1	50.5	28.7	68.8	26.6	61.1	15.6	330	0.512

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.1	95.3	79.1	93.9	79.1	92.6	78.9	81.6	89.6	81.0	88.9	80.4	88.2	17.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.5	153.1	84.0	79.1	151.2	84.1	78.5	148.0	84.2	45.2	89.5	44.5	88.8	43.9	88.6	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.3	24.3	22.0	DB	33.9	99.5	3.5	2.9	31.4	101.6	29.3	103.2	27.3	104.8	24.8	106.9	
			WB	30.8	82.7	3.1	1.2	28.6	83.6	26.7	84.3	24.9	84.9	22.7	85.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	75.2	61.0	64.7	69.9	75.5	80.7	84.5	85.3	85.9	82.9	77.5	69.7	63.7
	DBStd	10.67	8.72	9.13	7.31	5.63	4.16	2.64	2.22	2.38	3.29	6.04	8.05	8.62
	HDD50	38	18	9	2	0	0	0	0	0	0	0	2	7
	HDD65	523	179	110	42	7	0	0	0	0	0	4	49	132
	CDD50	9223	358	422	618	765	953	1034	1096	1114	987	853	593	430
	CDD65	4233	55	102	193	322	488	584	631	649	537	392	189	91
	CDH74	44433	211	422	869	2260	4935	7272	8120	8506	6261	3905	1250	422
	CDH80	17984	17	71	148	562	1645	3270	3850	4234	2611	1292	237	47
Wind	WSAvg	11.5	10.9	12.1	12.9	13.6	13.8	12.4	11.9	10.7	9.1	9.8	10.2	10.8
Precipitation	PrecAvg	25.0	1.0	1.1	1.3	1.6	2.0	2.5	1.9	2.0	5.4	3.6	1.8	1.4
	PrecMax	32.8	3.7	4.2	7.4	8.5	6.1	7.7	9.2	6.1	12.6	10.2	6.3	4.9
	PrecMin	14.3	0.0	0.0	0.1	0.0	0.0	0.4	0.1	0.0	1.1	0.0	0.0	0.1
	PrecStd	6.5	0.9	1.0	1.6	2.0	1.5	1.9	2.2	1.5	3.1	2.9	1.5	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.9	85.7	87.1	91.4	93.2	96.3	96.3	97.4	95.5	91.7	87.4	83.8
		MCWB	69.3	69.7	73.7	74.8	77.8	78.6	78.8	79.5	79.1	77.9	75.6	72.8
	2%	DB	79.0	82.1	83.5	88.1	91.0	93.5	94.4	95.5	93.3	90.0	84.8	81.1
		MCWB	68.5	71.2	72.0	74.2	77.4	79.1	79.0	79.3	79.1	77.6	74.0	72.2
	5%	DB	76.3	79.2	81.4	85.6	89.4	92.1	93.1	94.2	91.4	88.3	82.6	78.9
		MCWB	68.0	70.0	71.1	74.1	77.1	78.8	79.1	79.2	78.7	77.0	73.1	71.1
	10%	DB	73.4	76.1	79.0	83.2	87.5	90.7	91.7	92.6	89.9	86.5	80.8	75.9
		MCWB	66.6	68.7	70.3	73.2	76.4	78.6	79.0	79.1	78.4	75.8	72.5	70.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.9	74.0	75.2	78.7	80.5	82.4	82.0	82.4	81.9	80.8	78.3	75.6
		MCDB	77.0	80.7	82.1	85.4	87.5	91.5	89.8	90.1	90.1	86.5	83.0	79.4
	2%	WB	71.4	72.8	73.9	77.3	79.6	81.3	81.1	81.6	80.8	79.8	76.8	74.0
		MCDB	75.7	78.6	79.9	84.2	86.5	89.2	88.9	89.9	88.8	85.5	81.7	78.4
	5%	WB	70.0	71.5	73.0	76.1	78.6	80.4	80.4	80.8	80.2	78.9	75.5	72.7
		MCDB	74.2	76.2	78.5	82.4	85.3	88.1	88.2	89.3	88.0	84.8	80.3	77.1
	10%	WB	68.5	70.5	72.1	75.2	78.0	79.9	80.1	80.2	79.5	78.0	74.2	71.2
		MCDB	72.6	74.8	76.9	81.0	84.5	87.3	87.9	88.7	87.1	84.1	78.9	75.1
Mean Daily Temperature Range	5% DB	MDBR	16.5	15.4	14.6	13.9	12.9	13.0	13.2	14.1	13.2	15.0	15.8	16.1
		MCDBR	18.3	16.7	14.8	15.4	13.8	13.8	14.5	15.3	14.6	14.5	15.5	16.3
		MCWBR	10.5	7.8	6.6	6.3	4.4	4.0	3.8	3.9	4.8	6.0	7.6	9.2
	5% WB	MCDBR	15.6	14.1	13.1	13.0	12.0	12.5	13.1	14.3	13.0	12.8	12.9	14.3
MCWBR		10.4	7.7	6.6	6.0	4.4	4.4	4.3	4.5	4.9	6.4	7.3	9.4	
Clear-Sky Solar Irradiance	taub	0.369	0.382	0.401	0.448	0.464	0.434	0.454	0.439	0.443	0.396	0.380	0.372	
	taud	2.465	2.421	2.364	2.231	2.220	2.371	2.324	2.375	2.325	2.459	2.463	2.457	
	Ebn at Noon	279	284	284	272	266	272	266	271	267	276	273	272	
	Edh at Noon	31	35	39	45	46	39	41	39	40	33	31	30	
All-Sky Solar Radiation	RadAvg	979	1181	1463	1745	1945	2096	2083	1992	1656	1453	1109	884	
	RadStd	107	172	149	146	118	76	139	84	115	130	113	117	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+189

Nomenclature: See separate page